Robot Autonomy - Lab 3 - Task Planning

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Due Date: April 15

Introduction

This is an asynchronous lab assignment, which means that it is to be completed outside of lecture times and ultimately will involve running code on the robot. We encourage you to prepare as much as possible in simulation before you switch to the robot. This is also a group assignment, and the group will be the same throughout the rest of the semester. You only need to create one code base and video submission per group.

The purpose of this lab is to implement multi-step manipulation planning and to encourage the development of a reusable software stack. In particular, this lab focuses on combining task planning with multi-step action sequencing. We encourage students to reuse as much code from previous assignments and labs to build upon and complete this task.

NOTE - ROBOT EXECUTION AND SAFETY

Safety is paramount! Remember to always work with a partner (you are NOT authorized to work alone on the robots) and someone should have their hand on the emergency stop whenever the robot is running. Hit the stop immediately when anything happens that is even potentially close to unsafe for a human or the robot. Just because something works in simulation does NOT mean it will work exactly the same on the real system.

Robot Setup and Simulation

The robot will be setup in a shelf environment, with three shelves on its right and one on its left (robot's perspective). The shelves should not be modified. However, this lab does not rely on the shelves and can be performed solely on the table. We will setup the physical systems as close as possible to the simulated one, and the robots are shared between the groups. Altering the shelves will interfere with other teams' efforts.

You can use the simulation code from lab 2 to help you in testing your framework before deploying on the real robot. The task for this lab only uses the table and so it can be run on the real robot immediately.

Lab Task

To complete the lab, the robot should perform the following task. Start by arranging a set of blocks as shown in the first figure (4 red and 4 blue in illustration, but can be any color you choose). The nine gray squares represent a 3x3 grid of locations where the blocks can be placed. Important: the grid positions should be close enough to each that the fingers will collide with other blocks in neighboring squares, as illustrated in the bottom images. The robot should therefore select a suitable orientation for its top grasps of the blocks. The robot should also not drop blocks into an irrecoverable location (e.g., the robot should not drop a block into the middle of a ring of blocks).

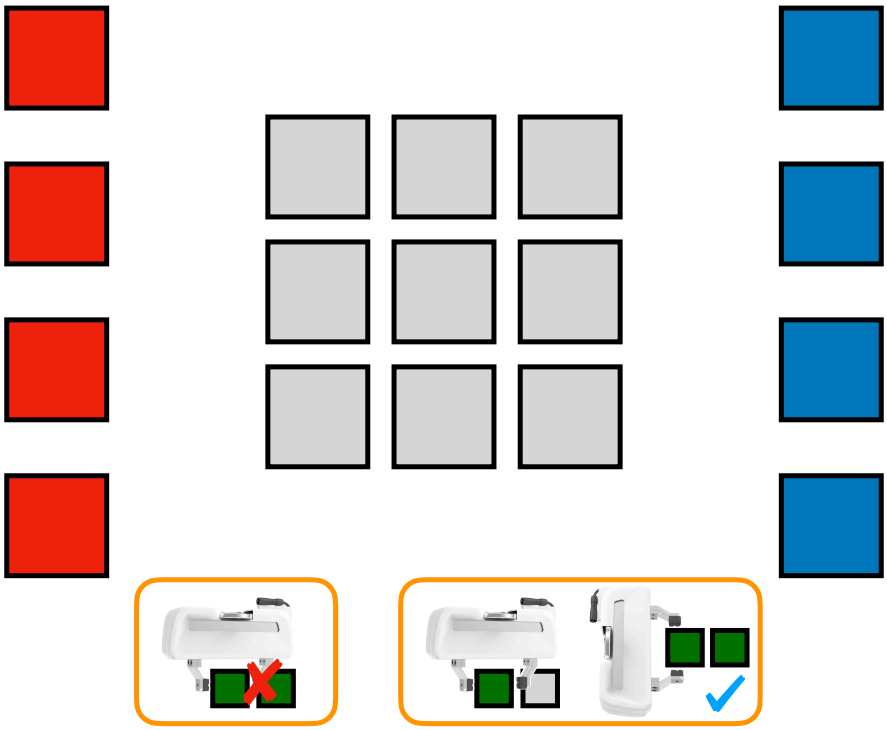

Figure 1: The initial setup with four blocks of different colors on each side and the 3x3 grid of block locations in the middle.

Beginning with this starting location, have the robot plan a sequence of actions to rearrange the blocks into the four patterns shown in figure 2. If no plan can be found for a pattern because it is infeasible, the robot should simply skip that step. The robot may directly reposition the blocks from one pattern to the next, or autonomously reset between each pattern.

Submission

For your submission, take a video of the robot performing the task. Please also include a copy of your code in the submission. You do not need to add narration or anything to the video, unless you want to. A simple video is sufficient to acquire full marks.

Note: During the first lab, some of the grippers were occasionally jammed during the pick and place task. If your robot ultimately has problems with the gripper, we will not deduct any points. You should still be able to run the overall behavior and collect a video. We will just assume that the gripper executed correctly, even if the block is dropped or the gripper never opened.

Disclaimer

This is the first time that we are running this lab. If you have any feedback or suggestions, or are experiencing any sources of frustration, please let us know as soon as possible on piazza. **These labs and the final project are meant to be fun!**

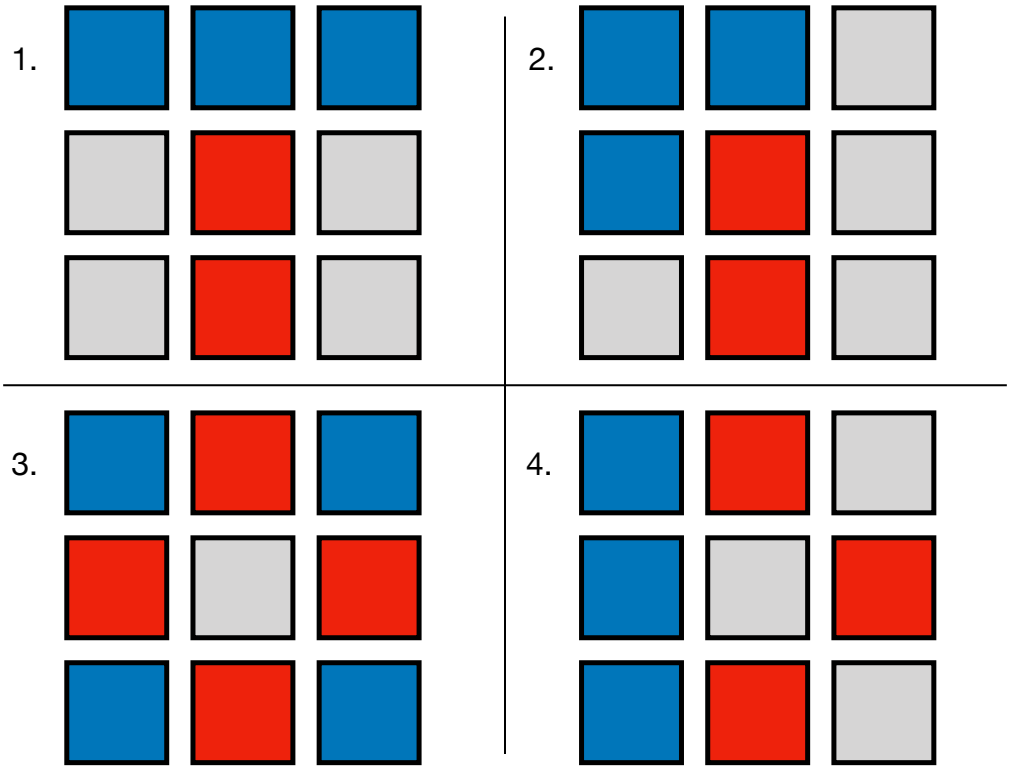

Figure 2: The sequence of patterns to create using the blocks. [pickaxe, axe, box, bow]

Acknowledgment

The MJCF model was adapted from the one made by Kevin Zakka and Saran Tunyasuvunakool. `https://github.com/deepmind/mujoco_menagerie/tree/main/franka_emika_panda`